

# **SuperMarket Sales Analysis Using Excel and Power BI**

## **Project Objective**

The objective of this project is to analyze supermarket sales data using Microsoft Excel and Power BI. The project focuses on cleaning, transforming, and analyzing raw sales data to identify business insights and improve decision-making. Excel is used for data preprocessing, while Power BI is used to create interactive dashboards and visualizations that help understand sales performance, customer behavior, product performance, and payment trends.

## **Problem Statement**

Supermarkets generate a large amount of sales data every day. Without proper analysis, it becomes difficult to identify high-performing products, profitable branches, customer purchasing patterns, and sales trends. This project aims to transform raw sales data into meaningful insights through data cleaning, analysis, and visualization.

## **Project Scope**

- Data collection from the supermarket sales dataset.
- Data cleaning using Microsoft Excel.
- Removing duplicates and handling missing values.
- Data transformation and formatting.
- Importing cleaned data into Power BI.
- Building interactive dashboards.
- Analyzing business insights from visualizations.

## **Dataset Description**

The SuperMarket Sales dataset contains transactional sales records from a supermarket. It includes information about customers, products, sales transactions, payment methods, and customer ratings. This dataset is used to analyze sales performance, customer purchasing behavior, branch performance, and product-wise revenue.

## **Power BI Data Modeling**

After completing data cleaning in Microsoft Excel, the cleaned dataset was imported into Microsoft Power BI. Data modeling is the process of preparing the data for analysis by ensuring that the dataset has the correct structure, data types, and relationships. A well-designed

data model improves report performance and enables accurate calculations and visualizations.

## **Create DAX Measures**

Examples:

- Total Sales
- Total Orders
- Total Quantity
- Average Sales
- Average Rating
- Gross Income
- Maximum Sales
- Minimum Sales

## **Dashboard Charts**

The dashboard was created in Microsoft Power BI to present supermarket sales data through interactive visualizations. Each chart highlights a different business insight and allows users to explore the data using slicers and filters.

- ✓ Card Chart
- ✓ Bar Chart
- ✓ Line Chart
- ✓ Column Chart
- ✓ Donut Chart

- ✓ Pie Chart
- ✓ Slicers
- ✓ Ribbon chart

## **Conclusion**

The SuperMarket Sales Analysis mini project was successfully completed using Microsoft Excel and Microsoft Power BI. The raw sales data was first cleaned and transformed in Excel by removing duplicate records, checking for missing values, standardizing data formats, and organizing the dataset for analysis. This process improved the quality and accuracy of the data.